

EUDICOTS

Almost three-quarters of the world’s known flowering plants are classified in the eudicot grouping, which evolved over 125 million years ago.

Eudicots get their name from the cotyledons, or seed-leaves, in the seed before it germinates. Unlike monocots, which have a single seed-leaf, eudicots have two. They include an enormous variety of plants, from agricultural weeds to towering rainforest trees, and they are of huge economic importance, as well as being prized as garden flowers. Many eudicots are annuals, with short lifespans measured in months or even weeks. Others are biennials and perennials, which live from two years to a century or more.

Despite their great diversity, eudicots have many physical features in common. Their leaves often have a net-like tracery of veins, unlike the parallel veins of monocots, and their stems have a well-developed vascular system, arranged in rings, for transporting water and sap. As well as getting taller, woody species thicken and strengthen their

stems as they grow. This “secondary growth” is absent in most monocots, which is why eudicots make up most of the world’s flowering shrubs and trees. Below ground level, most eudicots have a taproot, with smaller roots branching off it.

Eudicot flowers usually have parts in fours or fives, rather than the threes found in monocots, with sepals and petals that look different, both in colour and in shape. Each species has its own distinctive kind of pollen, but the grains always have three pores; monocot pollen grains only have a single opening.

A VITAL ROLE

The fossil pollen record suggests that the eudicots diverged from other flowering plants some 125 million years ago. Since then, they have colonized every habitat on land, although they are less common as waterplants. Their importance to animals is impossible to overestimate. Countless species – with the notable exception of grazing animals that feed on grasses (monocots) – depend on them for food and shelter, and in return, many act as pollinators, or help to spread their seeds.

DIVISION	ANGIOSPERMAE
CLADE	EUDICOTS
ORDERS	44
FAMILIES	312
SPECIES	Over 210,000

DEBATE FOUR-WAY SPLIT

For many years, all flowering plants were classified in two groups – the monocotyledons (monocots) and the dicotyledons (dicots) – based on the number of their seed-leaves. However, DNA analysis, combined with palynology – the study of pollen – has shown this classification does not fully reflect plant evolution. As a result, flowering plants are now split into four overall groups: basal angiosperms, magnoliids, monocots, and the eudicots, or true dicots.

NEW PERSPECTIVES

Computer analysis of DNA data has suggested new ways of subdividing the eudicots. These new hierarchical groupings, known as clades (see pp.30–31), give insights into the relationships between species and how they evolved, but they are difficult to interpret in terms of physical characteristics of the resulting groups and do not fit neatly into the traditional taxonomic hierarchy. This book is organized around orders and families, which still provide the clearest way to understand the diversity of flowering plants. However, the table below shows how they “nest” together in the proposed new system of clades.

EUDICOTS (pp.150–209)
(BUXALES – DIPSACALES)

CORE EUDICOTS (pp.155–209)
(GUNNERALES – DIPSACALES)

SUPERROSIDS (pp.156–181)
(SAXIFRAGALES – SAPINDALES)

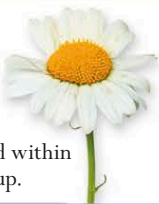
ROSIDS (pp.158–181)
(VITALES – SAPINDALES)



As its name suggests, Japanese rose belongs to the rosid group.

SUPERASTERIDS (pp.182–209)
(CARYOPHYLLALES – DIPSACALES)

ASTERIDS (pp.192–209)
(CORNALES – DIPSACALES)



Oxeye daisy is an asterid within the larger superasterid group.

BUXALES

Containing just a single family of around 120 species, the order Buxales is found in temperate, subtropical, and tropical regions. Most of the members are trees or shrubs, with simple, evergreen leaves and separate male and female flowers on the same plant. Many species are grown ornamentally and box wood is used for carving.



SWEET BOX
Sarcococca hookeriana
F: Buxaceae

This species bears clusters of small fragrant flowers in winter, and is found in shady habitats in W. China.



COMMON BOX
Buxus sempervirens
F: Buxaceae

Native to rocky woods and scrub from Europe to N. Africa, this evergreen shrub or small tree is often grown in gardens for topiary.